Computing Systems and Programming (1): Intro

Dr. Clifford Choy

School of Design

16 Oct 2026

Where can we find computers?






Friday, Jun 7 | 86°F


kindle
kindle



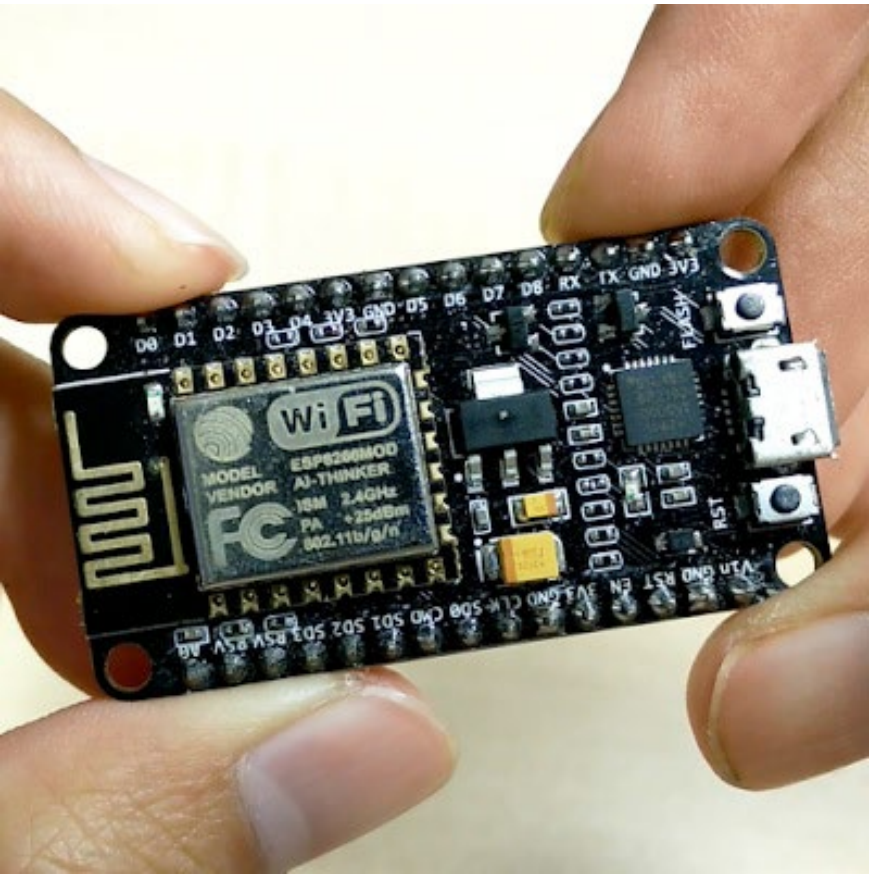


Bandi Tamagotchi





Nintendo Game & Watch



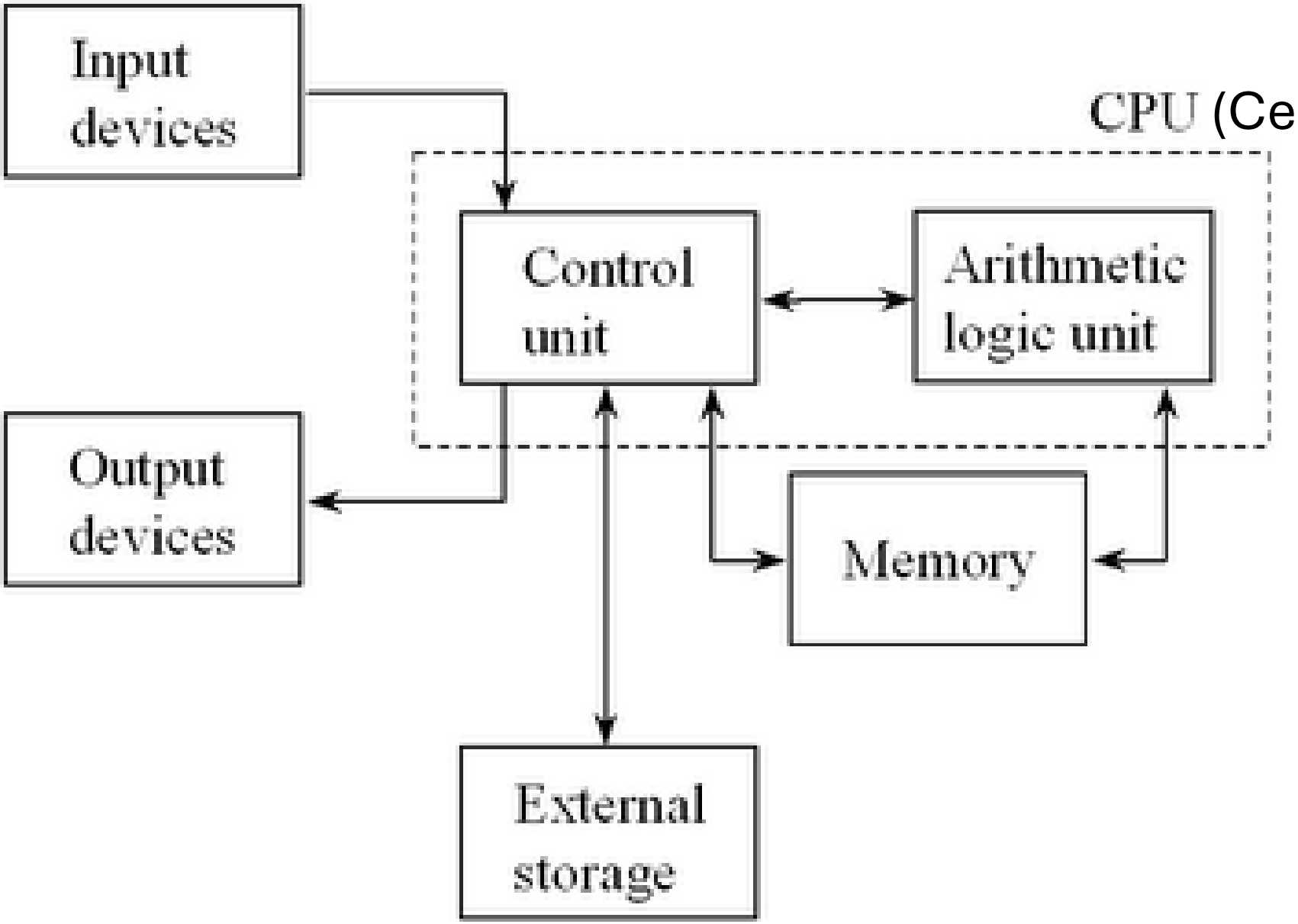
NodeMCU v1.0



Xiaomi Mijia Smart Sensor Set



Octopus Card

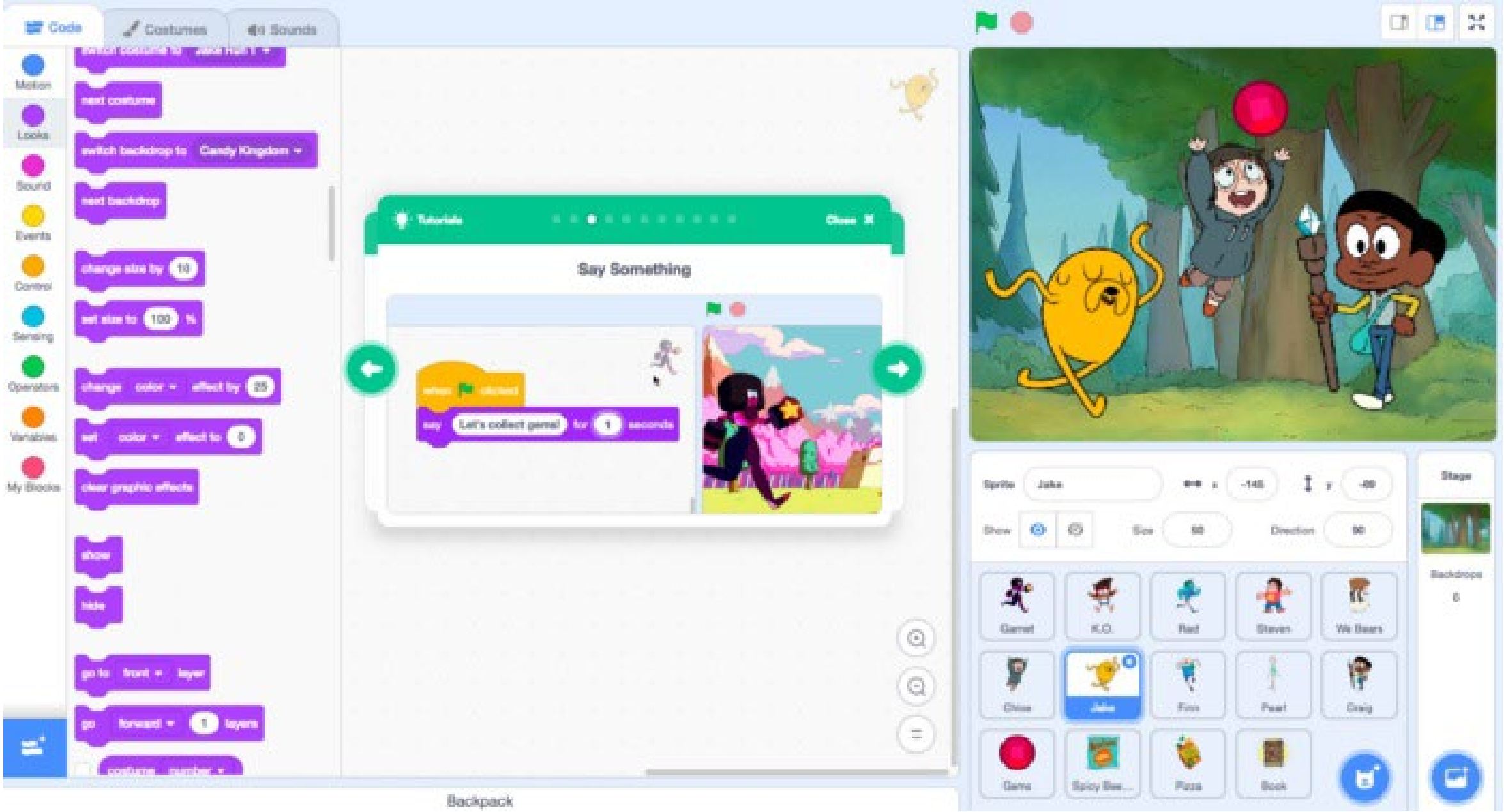


The von-Neumann Architecture. http://radialmind.blogspot.com/2010/09/von-neumann-machine-parallel-processing.html

What is a Program?

- A sequence of instructions for computer to follow to perform some useful tasks
- Think of it as a recipe or instruction manual, but for computer

- The program instructs the computer to perform useful tasks through interfacing with its **Application Programming Interface** (API)
- Computer cannot tolerate errors/uncertainties; hence a poorly written program results in poorly ”behaved” system/product



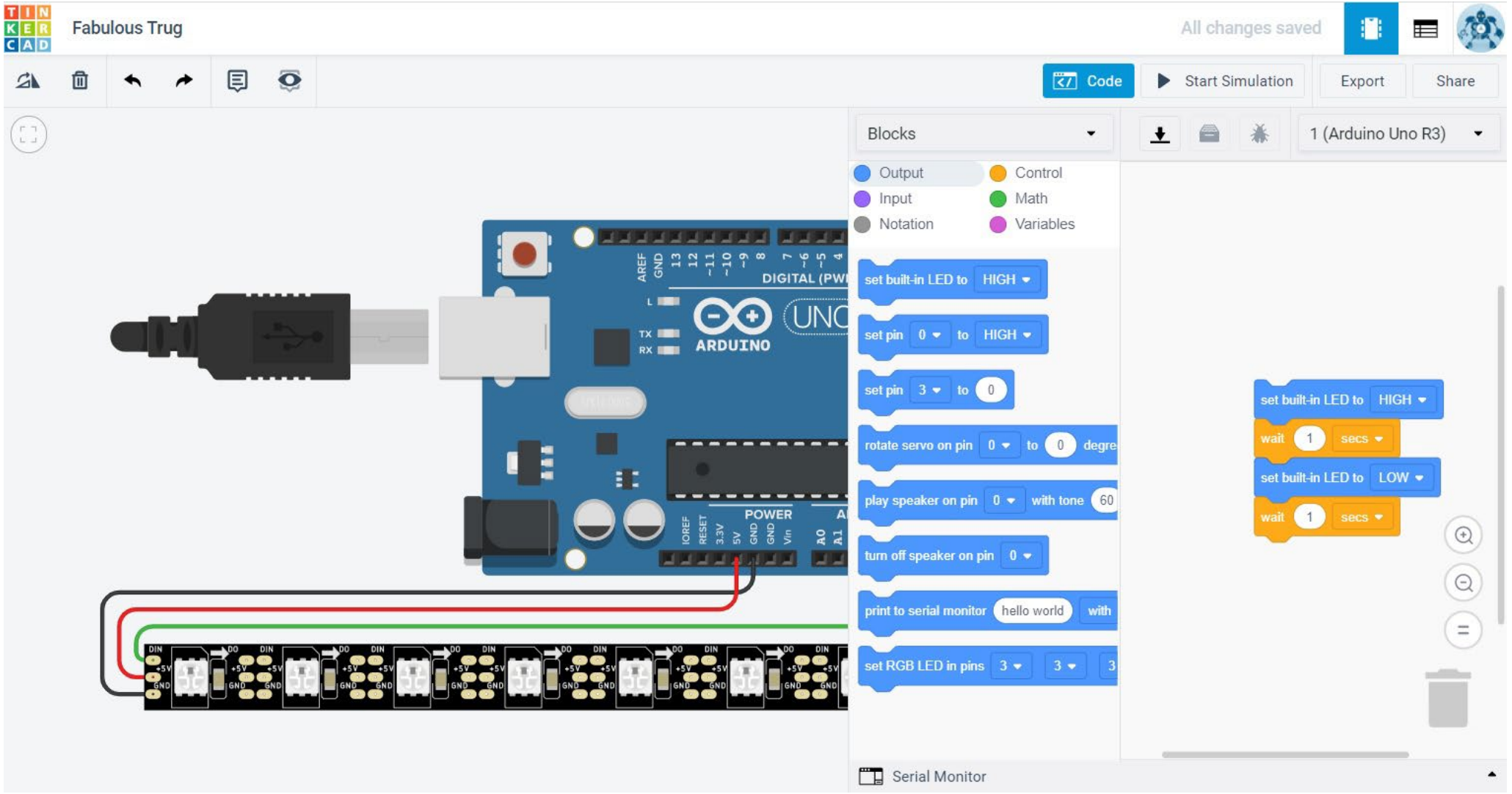
https://medium.com/scratchteam-blog/introducing-scratch-3-0-expanding-the-creative-possibilities-of-coding-ad4cd9eda9ef



Build Your Own Android Apps With MIT App Inventor

https://geekdad.com/2015/02/mit-app-inventor/

Tinkercad Circuits



Arduino (C++) (Embedded Systems Programming)



iocontrol_serialout | Arduino IDE 2.3.6

File Edit Sketch Tools Help

AirM2M_CORE_ESP32C3

iocontrol_serialout.ino

```
1  //
2  //  for demonstration use of handler function in IOControl for simple printing on Serial Monitor
3  //    circuit: use Luatos ESP32C3 board
4  //
5
6  #include <IO_Control.h>
7
8  DeclareHandler(hdlr,_ctrl,IOControl,{ //  declare a handler for scheduler
9    //  hdlr - name of the handler function
10   //  _ctrl - Control object
11   //  IOControl - class of object of this handler
12   //  _currentMillis (ulong_t) - current time when this handler is invoked
13   //  _status (int) - a number indicating status of this handler
14
15   // if (_ctrl.isSchedulerTimeEvent(_sttus))
16   //     Serial..println("IOControl - SchedulerTimerEvent");
17   Serial.print("current time=");
18   Serial.println(_ctrl.handlerCurrentMillis);
19 });
20
21 void setup()
22 {
23   // put your setup code here, to run once:
24   IO_Control.begin();
25
26   Serial.begin(115200);
27   Serial.println("chip started");
28
29   IO_Control.enableScheduler(0, 100, hdlr);
```

AirM2M_CORE_ESP32C3 on COM8 [not connected] 1

HTML/Javascript (Web Programming)

```
<html>
<body>

<p id="status">
</p>

<p> Send msg from JS
<input type="text" id="js_msg_send">
<button onclick="sendToAI2_from_DOM(js_msg_send,true);"> send (ASAP) </button>
<button onclick="sendToAI2_from_DOM(js_msg_send,false);"> send (Wait for timeout) </button>
</p>

<p> Send msg from AI2 (simulated)
<input type="text" id="ai2_msg_send">
<button id="ai2_msg_send_button" onclick="sendFromAI2_from_DOM(ai2_msg_send);"> send </button>
</p>

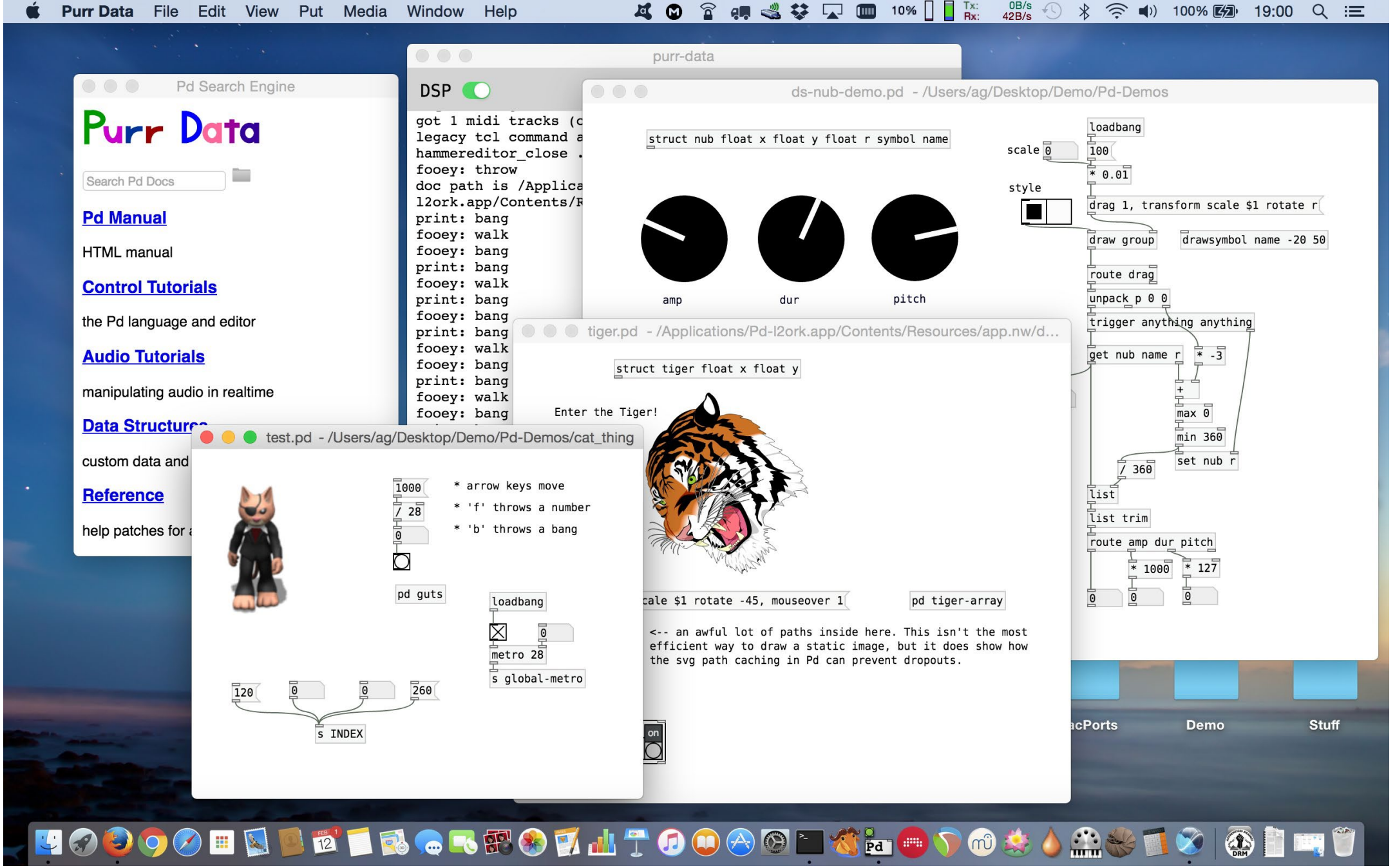
<p> Msg from AI2
<input type="text" id="ai2_msg_recv" disabled>
</p>

<script src="https://code.jquery.com/jquery-3.4.1.min.js"></script>
<script>

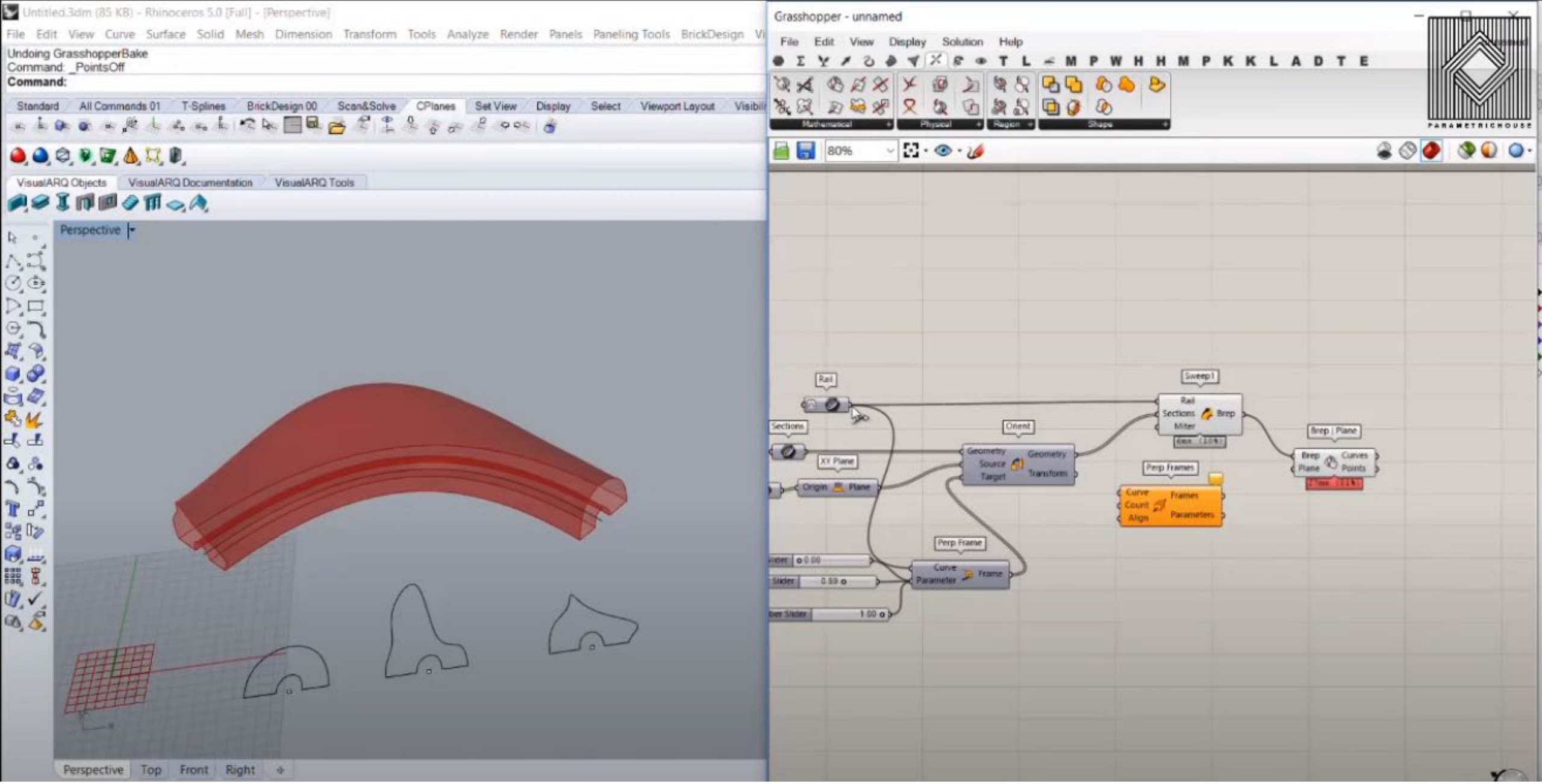
var flowState={
    AI2PollTimeMS:      100,    //  time between poll to check if new msg from AI2 available from WebViewString
    JSMaxReturnTimeMS:  1000,   //  maximum time for JS to send out msg
    JS2AI2:             true,
    nextSeqNum:         0,
    sendCallback:       [],
    sendObj:            [],
    sendASAP:           false,
    jsTimer:            null,   //  timer for sending from JS
    msgHandler:         ( obj ) =>
                        {   ai2_msg_recv.value=JSON.stringify(obj);
                            //console.log("obj from AI2:"+JSON.stringify(obj));

                            if (obj.hasOwnProperty("string"))
                            {   let return_obj = {str: btoa(obj.string)};
                                sendToAI2(return_obj,true, ()=>{js_msg_send.value=JSON.stringify(return_obj
                                );});
                            } else
                            {   //  nothing return to AI2
```

Hyper Text Markup Language file | length : 4,397 lines : 150 | Ln : 75 Col : 5 Sel : 0 | 0 | Windows (CR LF) | UTF-8 | INS



https://agraef.github.io/purr-data/

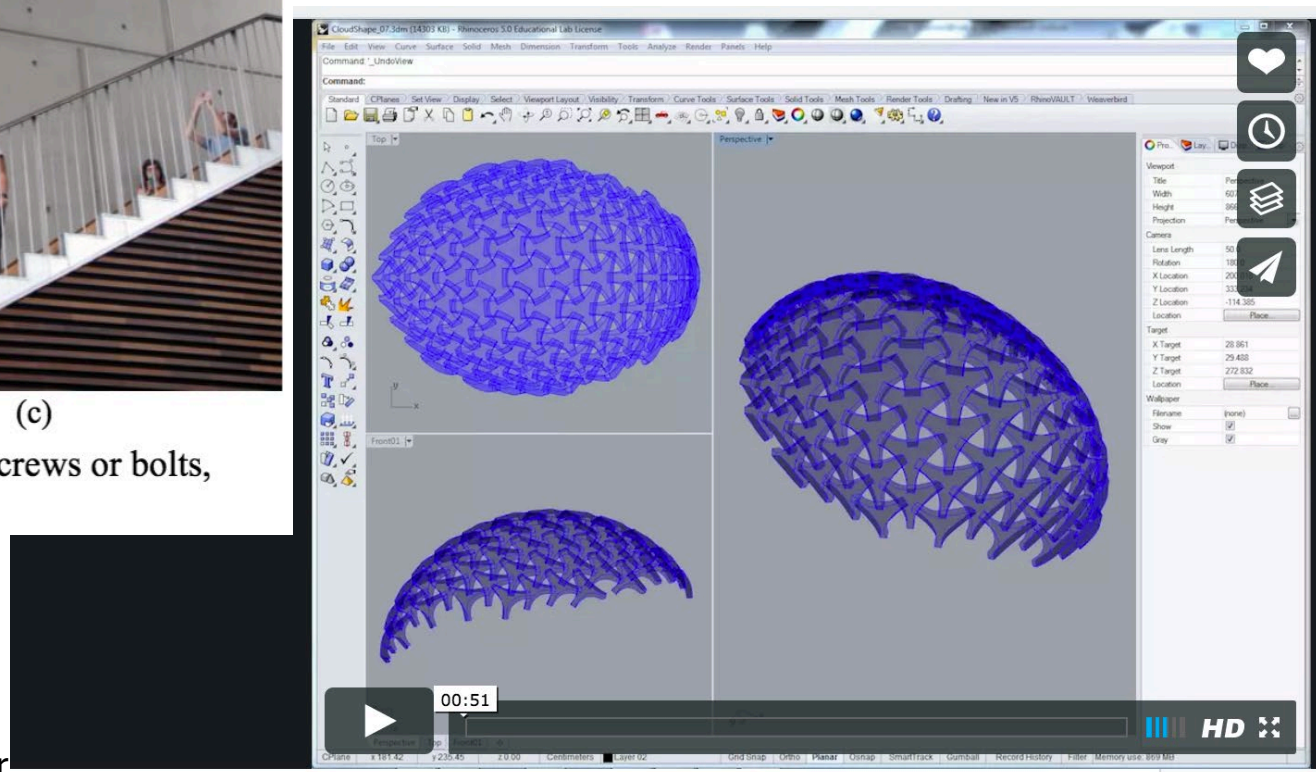


https://youtu.be/aQ0UIOf_50E



(a) (b) (c)

Figure 1: (a) Compact, flat-stacked configuration, (b) rapid assembly without screws or bolts, (c) Undulatus pavilion hanging at the IASS2015 expo.



Brancart, S., VERGAUWEN, L., ROOVERS, K., VAN DEN BREMT, D., DE LAET, L., DE TEMMERMAN, N. (2015) UNDULATUS: design and fabrication of a self-interlocking modular shell structure based on curved-line folding. Proceedings of the International Association for Shell and Spatial Structures (IASS)Symposium 2015, Amsterdam Future Visions17 -20 August 2015, Amsterdam, The Netherlands.

UNDULATUS pavilion (IASS 2015 expo) - Demonstration of 3d modelling in Rhino and Grasshopper

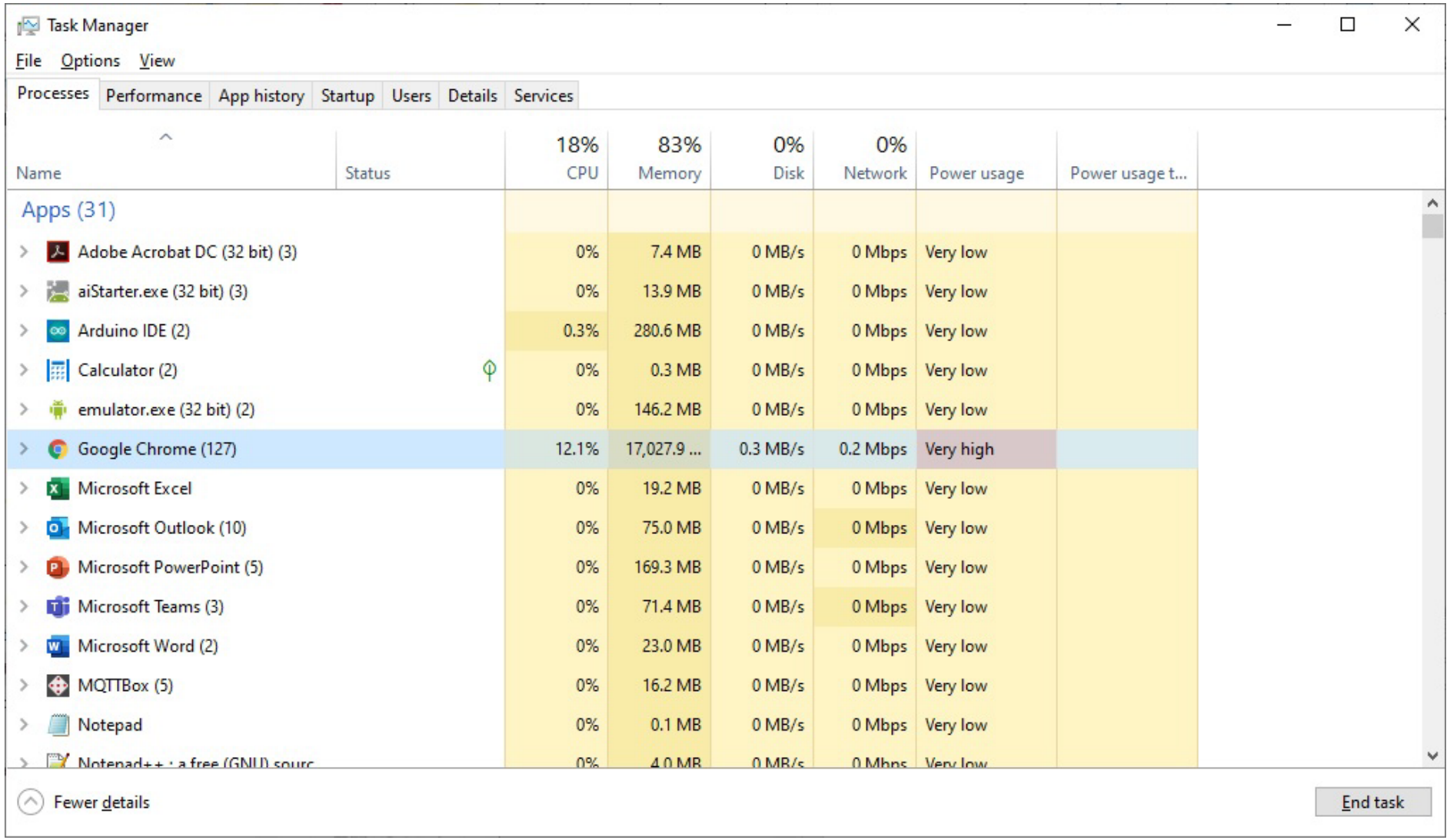
from **VUB ARCH** 1 year ago | more

Related Vi

Autoplay

https://vimeo.com/137466525

Processes (1)


Task Manager
File Options View
Processes Performance App history Startup Users Details Services
Name Status 18% CPU 83% Memory 0% Disk 0% Network Power usage Power usage t...
Apps (31)
Adobe Acrobat DC (32 bit) (3) 0% 7.4 MB 0 MB/s 0 Mbps Very low
aiStarter.exe (32 bit) (3) 0% 13.9 MB 0 MB/s 0 Mbps Very low
Arduino IDE (2) 0.3% 280.6 MB 0 MB/s 0 Mbps Very low
Calculator (2) 0% 0.3 MB 0 MB/s 0 Mbps Very low
emulator.exe (32 bit) (2) 0% 146.2 MB 0 MB/s 0 Mbps Very low
Google Chrome (127) 12.1% 17,027.9 ... 0.3 MB/s 0.2 Mbps Very high
Microsoft Excel 0% 19.2 MB 0 MB/s 0 Mbps Very low
Microsoft Outlook (10) 0% 75.0 MB 0 MB/s 0 Mbps Very low
Microsoft PowerPoint (5) 0% 169.3 MB 0 MB/s 0 Mbps Very low
Microsoft Teams (3) 0% 71.4 MB 0 MB/s 0 Mbps Very low
Microsoft Word (2) 0% 23.0 MB 0 MB/s 0 Mbps Very low
MQTTBox (5) 0% 16.2 MB 0 MB/s 0 Mbps Very low
Notepad 0% 0.1 MB 0 MB/s 0 Mbps Very low
Fewer details
End task

Processes (2)

- A running program
- Has its own “world” with its own set of data (**variables**) independent (usually) of other processes of the same program
- Starts off with nothing inside, and instructions in the program tells computer what to do in this “world” for manipulating those data inside the “world” to achieve some meaningful goals
- Note:
  - “Paint” in Windows can be run for more than once, each with its own data (the image to be displayed)
  - “Chrome” can be run for more than once; however, “world” in each of these processes are somehow connected
  - Some program can only have one running process, e.g. the operating system (Windows, MacOSX, Android)

Programming in Arduino

Concepts

- Arduino IDE and IO Control library
- Functions and Libraries
- Data Types, Variables and Operators
- Conditionals
- Loops
- Arrays

- UML Activity Diagram

IO Control library

- A library developed in PolyU Design to support “scheduled events”
  - https://www.designandmake.org/x/UQHfE
- Installation
  - Refer to “Installation of Third Party Libraries” in https://www.designandmake.org/x/FIDhEQ to install all necessary libraries to use the IO Control library
  - Download the latest version of IO Control library and install it via “Sketch > include Library > Add Zip Library …”
- Compile “IDDBoardTest_2025” and upload to ESP32C3
  - Open “IDDBoardTest_2025” from Examples
  - Select “Tools > Partition Scheme > Huge APP”
  - Click “Upload” button
- Install Notepad++ as Arduino IDE is not too “programmer friendly”
  - https://notepad-plus-plus.org/

- Community
- Arduino
  - ESP32
    - LittleFS Initialization (ESP32)
    - ESP32 C3 (LuatOS)
      - Arduino IDE Installation for ESF
      - **Solutions to Macbook with M**
      - How to Enable Personal Hotspc
      - Solution for ESP32C3 not show
    - ESP32 S3
    - ESP32 LittleFS Uploader
  - ESP8266
  - IO_Control Library
- Interactive Desktop Decoration
- 3D Modeling
- 3D Printing
- File
- Mechanisms
- Parametric Design
- Recruitments
- Tutorial
- Tutorial - Shared Links

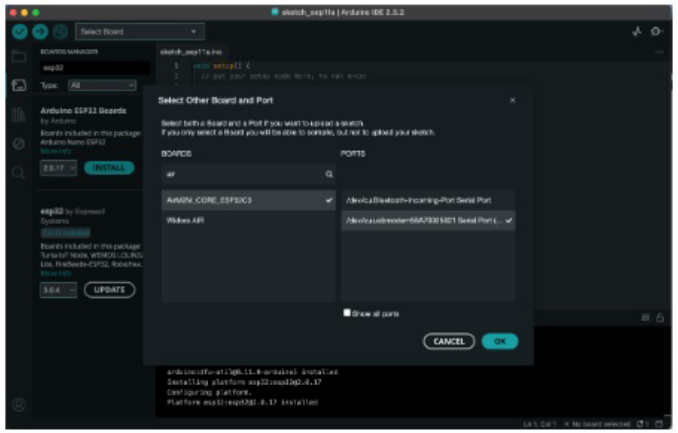
Space tools

Solutions to Macbook with M1 Chip encounters error when uploading program to ESP32C3

Created by Ho Yee Tung on Mar 26, 2025 18:29

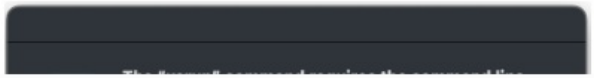
- **Problem 1: Error message mentioned "xcode-select: note: No developer tools were found, requesting install"**
- **Problem 2: Error message mentioned "A fatal error occured: Failed to write to target RAM (result was 0107000: Operation timed out)"**

Note 1: Make sure you choose the "correct board" and "correct port" just like the following image if you see any red words (error) shown in the output section.



Note 2: See if there's a prompt message pop up about "The xcrun command requires the command line developer tools"

Please install the software as follow.



https://www.designandmake.org/x/QyyFDw

Learning Arduino Programming (1)

- Arduino programming
  - C++ programming language
  - Arduino specific libraries

- Arduino is using C++ programming language, although it called itself “Wiring” (https://docs.arduino.cc/language-reference/)

- Many libraries for Arduino programming, each for using specific hardware functions
  - E.g. if you want to use SSD1306 (OLED display), you will need “Adafruit SSD1306” library

Learning Arduino Programming (3)

- https://docs.arduino.cc/programming/
  - For conventional programming on Arduino, and basic language syntax
  - Does not use IO Control library, and hence cannot support scheduled events
- https://www.w3schools.com/cpp/default.asp
  - On C++ programming language

Preferred Order in Going Thru Examples (1)

- Core functions/methods for all “Control” classes
  - iocontrol/iocontrol_serialout
  - iocontrol/iocontrol_serialout_fsm1, iocontrol/iocontrol_serialout_fsm2
  - iocontrol/iocontrol_serialin
  - iocontrol/Iocontrol_setTimeoutByRelTime, iocontrol/iocontrol_fsm_setTimeout_serialin, iocontrol/iocontrol_setTimeout_wc
- GPIO (outputs and inputs)
  - gpio/binout/binout_timedwrite
  - gpio/binout/binout_serialin_timedwrite
  - gpio/binout/binout_play_oneseq
  - gpio/binout/binout_play_altseq
  - gpio/binout/binout_serialin_playseq_immed
  - gpio/binout/binout_serialin_playseq_follow
  - gpio/binin/binin_changetime

Preferred Order in Going Thru Examples (2)

- ADC
  - adc/adc_read
  - adc/adc_thres
  - adc/adc_binin_joystick
- PWM
  - pwm/pwm_setduty
  - pwm/pwm_play_oneseq
- Servo motor
  - motor/servo/servo_setAngle
  - motor/servo/servo_play_oneseq
- Tone
  - tone/tone_setTone
  - tone/tone_play_oneseq